



Course Video One-Way ANOVA Transcript

Hello everyone. Welcome to lesson 1 video 2. In this video, we will study one-way ANOVA. This is the simplest form of analysis of variance, where the comparison is between 3 or more than 3 groups based on one factor.

We learned how to set up good experiments, but today we are going to look at the tool we use to analyse them. We will expose the difference between one-way ANOVA and two-way ANOVA and we will learn about how to make sense of the results. So, what exactly is one-way ANOVA? One-way ANOVA is when you have one factor.

Two-way ANOVA is when you have two factors. Let's say if you are trying to study whether 3 different teaching methods has any effect on the exam scores of students. Now, you might have 3 different teaching methods as method A which could be lecture based, method B which could be audio-video classes based, lecture C which could be interactive session based.

But you have one categorical variable which is teaching method. It has 3 different categories, the 3 types of methods. Let's say we call it as method A, B, C. Now, we would want to understand whether a student's score is at all dependent on these 3 different types of methods that we employed to teach them.

So, that simply is one-way ANOVA because you have one factor which is teaching method. Now, think of what could be two-way ANOVA when you have two factors. Now, in this particular case, let's say you want not only to analyse how does student's marks changes based on the 3 different types of teaching methods we employ on them versus we also want to study the different types of assignments that we give.

So, assignment could be of type 1 which could be experiential learning assignment, the assignment could be very rigorous theoretical assignment, the assignment could be a reading experiment. So, you give them something to read on and that could be different types of assignment. So, now the factor for study is either one variable which is one-way ANOVA or it is two variable which in this case is the different type of teaching method which is one variable and the different types of assignments given which is two variable.

So, you have the choice of one-way ANOVA versus two-way ANOVA or you can have many-way ANOVA if you have many number of factors, but the core concept still is the same. You have a categorical variable or combination of categorical variables. You are trying to analyse how those variables either one or a combination affects the output.

In this case, the output or response is the marks that students score. So, let's learn a little more deeper about all of these things. What is one-way ANOVA? One-way ANOVA is when you are comparing multiple groups and you have only one factor.

So, you have teaching methods. Let's say we want to understand how does one different type of teaching methods affect the marks scored by students versus the other



type of teaching method. So, in this case, let's say the factor is teaching method, but the levels of the factors are lecture, group project and interactive study.

And the response variable is the exam scores that the students get. So, you would frame the hypothesis as all the groups are same or at least one group is different. It could very well be let's say people or students who went through interactive study, they scored much better than people who went through lecture based studies or group based project.

So, you could have a situation wherein all the groups are same. There is no significant difference which is your null hypothesis or you could also have a situation wherein you have one group which is different from other group or you have all the groups which are different. So, you say that at least one is different and it means that there is a difference which exists, there is significant difference which exists, but it is within one or it could be two or it could be three or if you have more number of levels, then you would say that those many differences exist.

Now, if you think of ANOVA doesn't tell you whether which groups are different. It only tells you if your alternate hypothesis is true, it tells you that at least one group is different. It doesn't tell you which group is different.

So, what you'll have to do is after you conduct your analysis of variance, you go back again and run multiple t-tests to test which particular group is different. So, you will take two groups and then you will start analysing them. You will see if the two groups are same or different.

Then you will move on, choose two different groups again and then you would do as many experiments required for comparing two groups dependent on how many groups do you have. So, if you have n groups, you could have n choose two combination of ttests. This is the results of one-way ANOVA and this is how we structure the output of one-way ANOVA.

Here in the source, you have variance coming from the groups, variance coming from the error. The total variance degrees of freedom is nothing but total amount of groups minus one. Here you have total, which is total number of data points.

let's say we have three different groups. Each group has 10 10 data points. So, 10 plus 10 plus 10 is 30 and 30 minus 1 is 29, which is what you have listed here.

Now, sum of squares is how much variance is between the groups. If you see this row and it is 540.8. Now, if you see the bottom, it is within the group. The variance within the group is 775.2. Now, you could sense that the variance between the group, if you think of variance between the group is small, but if you divide sum of squares with degrees of freedom, it becomes 270.4, which becomes your mean squared error.

Now, when you divide with sum of squares of errors with respect to the total number of degrees of errors, degrees of freedoms that you have for errors, it becomes



eventually small. Now, this is what we use to compute the F statistic. In this particular case, if you think F statistics, for F statistics, the numerator is 270.4, the denominator is 28.7. Hence, F value would be higher.

Obviously, if the variance between the group is higher, the mean variance between the group is higher, then the mean variance within the group, then F statistics is higher. It means that there is a statistical significant difference between the groups. You again look for P value, which is what highlighted as red here.

And if that P value is less than the level of significance, which we generally take at 5%, so if P value is less than 0.05, you would go ahead and say that the alternate hypothesis is true. I hope this explanation was clear to everyone. But what you also should look for is whenever you have been given a problem or you are solving a problem, whether an exam or an industry, you have to create a structure like this, you have to create an ANOVA table like this.

And then if your P value comes out to be lesser than level of significance, or you have high enough F statistic value, then you say that your alternate hypothesis is true. Now, two-way ANOVA, as I have told you, there are two factors. This line states that.

You have two different factors. You have levels of factor A, and you have levels of factor B. And then you also study for the interaction between the two factors. Okay.

So, if you think of, does factor A has any effect? This is what the first line measures. Does factor B has any effect? Okay, that's what the second line is. And in the third line, we are saying more interesting thing.

We are saying that are the two factors interacting with each other, which means that both has the effect on the output or the experimentation that you're doing. Okay. So, you could have a situation wherein you have effect only coming from one variable, effect only coming from two variables, or effect coming from both of them combined.

That is what is called as interaction effect. Okay. Now, this is a simulation which I set up for all of us to understand how does the interaction term comes in, or it can be visualised.

On the y-axis, you have response variable. You have factor level A, which has two different levels, if you think of. You have factor level B, or let's say factor level A, which has two different levels here.

Now, factor B levels are also here. This is level one, this is level two factor, this is level one of factor A, this is level two of factor A. Now, in this particular case, when you have no interaction, you would see that on the response variable, there would be the change. Let's say, if there is a change because of the change in factor level of the first variable or the first factor, it changes from level one to level two, the response variable also changes.

But you see the same effect happening on response variable when you choose factor level B as well. So, that's what is denoted here, if you think of. Now, in the cases when you will have interaction, you will see that the increase in one causes, let's say, when one of the factor here, factor B, it changes from one particular value to another value in terms of your response variable.

The other variable also does changes. So, you would get a cross sign like this and the lines do not remain parallel. So, an interaction is present when the relationship between one factor and the response variable changes depending on the level of other factor.

Based on whatever level you are for the other factor, your value of the response variable changes with respect to the first factor. That's what the interaction effect is. So, the main guiding principle is, if you have parallel lines, there is no interaction.

And if you have intersecting lines, it means that the interaction is present. Now, this is again the same example that we took fertiliser and plant variety. We have two different types of fertilisers, fertiliser A and fertiliser B. And we have two different types of plants, plant X and plant Y. Now, we want to study in this simple experiment, whether the two different types of plants are having an effect on the sizes or it is because of the fertiliser.

So, you see here, you have set up the entire table again. You have source, which is fertiliser or variety or interaction. Now, you have degrees of freedom 1, 1, 1. You have F statistics that you have computed.

Now, if you see that the F statistics for individual effects, the individual effects, which is the effect coming from factor A and the effect coming from factor B, they are very, very small. The F values are small. But the effect coming from interaction term is really high, which means that the P value is going to be lesser than 0.05 or the level of significance.

And hence, you conclude that it is because of the variety of the plant and the fertiliser, which causes the effect. So, the fertiliser, what does this line mean? Think of it as there could be different type of plant on which the fertilisers are showing effect. So, this is all what you have in two-way ANOVA.

I hope this was clear for you to make the concept more clear. Similarly, you have this second example. I will not go through it because I think you have already gone or you have understood the effect.

Herein, you have the sort of two different types of factors. One is study method, which is visual and audio, and then different times that you have, times of the day, which is morning and evening. And you are trying to study the effect of a study method along with times of the day to understand whether there is an effect on the marks scored by students or test scores of the students based on these two factors and the interaction between them.

Now, in this case, if you see, the interaction term value or the status value is not very high. But the p-value for one of the methods or one of the factors is very high, very low, which means that study method is important. Which type of methods do you use, whether it is visual or auditory? It is the main effect.

It is what is causing the difference. So, generally, in two-way ANOVA, you will set up the experiments to differentiate between the main effects, which are these two, versus the interaction effect. If all of them are important, you will see that the p-values are all significant here, which is less than 0.05. Now, in this particular case, you see that only the first one is significant, which is the study method is making the difference.

So, in summary, what we have learned, this is an important topic. So, I will try to summarise it a little by taking a little more time. We went through analysis of variance, wherein we are saying, if we would want to understand the effect of a categorical variable, a categorical variable is something which has categories, right, or a variable which has categories is called as categorical variable.

You could have a teaching method, which are teaching method A, B, C, D, E, F, and so on, right, different levels of teaching method. So, that is a categorical variable. And when we are trying to study the effect of categorical variables on a response variable, here in the examples that we took, the response variable in the first case was the plant height.

And in the second case, we were looking into the test scores of students, right. So, what effect does the categorical variable either if there are one variables, or there are multiple types of variables, right, or factors, what effect does that categorical variable have on my response, when we are trying to study that, then we are using one way or two way ANOVA. If you have one variable, or one factor, we call it as one way ANOVA.

If we have two factors, we call it as two way ANOVA. The purpose in one way ANOVA is to compare the means of three plus groups or let's say, when I say groups, groups as in there is only one factor, but divided among multiple levels, right. So, these are the groups, teaching method A, B versus C. So, if you have teaching method A, which is audio based, teaching method B, which is video based, teaching method C, which is both audio video based.

So, you have three different groups, right. The hypothesis would be whether all groups are equal, or there is no difference between the groups, or there is at least one group which is different. That's what we assumed if you remember, right.

And then we do the test and come up with the output. In two way ANOVA, we have two different factors or two different categorical variables, right. And then we are not only studying the main effect, which are the effect of two variables, but also the interaction effect, right.



And that's what we looked into. So, I hope all of this now starts making sense to all of us and understanding how do we measure the effect of factors, which we call as categorical variables and their levels on any response variable, right. When you have multiple types of groups or multiple groups, and you would want to understand whether those groups are having any statistical significant effect on the response variable that you are looking at, right.

Let me tell you a very simple and convenient example. Let's say Maruti Suzuki has three different production plants in India. And those production plants are in Pune, Bangalore, Hyderabad.

Now, if you would want to see are the mileage of cars equal for the cars being produced in Pune, Hyderabad and whatever the third unit is, let's say Bangalore, right. So, we might want to do a test. Now, you have the mileage of the cars, which is the response variable that you are using.

And you are saying whether the mileages are different in three different groups. Groups are your cities, which are Bangalore, Pune and Hyderabad, right. So, you would want to understand whether the cars being produced at Bangalore, Pune and Hyderabad, are they same based on their mileage, or are they different, right.

Similarly, different types of people living at different places, are their heights equal or they are different. You can think of multiple types of groups now that you understand the groups and how can we use one way ANOVA to differentiate between groups. Now, in this particular case, when I said height of groups, you have only one factor, which is height.

But people at different places are levels, let's say people at Jammu and Kashmir and Uttar Pradesh and Uttarakhand and Kerala and Karnataka and so on, right. So, you have different levels. Now, you would want to compare whether the heights are actually same or different among different groups, right.

So, that is what we use. Remember, again, letting you know that you cannot directly come up with which particular groups are different in ANOVA. It just tells you, are there differentiating groups or are there groups which differ from each other or not.

That's all. It will not tell you which particular group is different. Now, after you conduct F-test or analysis of variance, you will go again back to your t-test, compare two groups and then say whether which one of them are having the difference, right.

I hope this was clear to everyone. In summary, one way ANOVA provides a framework to test whether three or more than three groups, their means are equal or not. By comparing between group and group means or between group variances and within group variances, we can determine if the observed differences are statistically significant or not.



This forms the foundation or more foundation for advanced analysis that we can do in the next set of ANOVA studies that we have. Thank you.

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